

02/23/2026

More on higher categories

$\mathbb{S}$ (commutative) monoids

Last time The homotopy hypothesis, underpinnings of higher category theory

Today Some axiomatics of higher categories, (commutative) monoids.

Axiomatics

- > Even though we won't go through the axiomatics in detail, we'll start using these tools and hopefully demonstrate how to work with the axioms.

Notation. We'll write Cat_∞ for the ∞ -category of ∞ -categories.

Def. An ∞ -category $\mathcal{C}$ is an anima (or ∞ -groupoid or space) if every map in $\mathcal{C}$ is invertible, i.e., for every map $f: x \rightarrow y$, there exist maps $g: y \rightarrow x$ and $h: y \rightarrow x$ along with equivalences $gf \simeq \text{id}_x$ and $fh \simeq \text{id}_y$. We write $\text{An} \hookrightarrow \text{Cat}_\infty$ for the full subcategory spanned by the anima.

Remark. Inspired by Beilinson, the term 'anima' was introduced by Clausen and Scholze. The point is that one wants a term for 'homotopy types' that does not refer to any specific model of homotopy theory. The term is Latin for 'soul'.

In Sheaf theory, we'll often want to consider topological spaces as genuine topological spaces, and not models for homotopy types. So it is convenient to have a (short) term that distinguishes them.

Axiom For each ∞ -category $\mathcal{C}$, there exists a natural functor

$$\mathrm{Map}_{\mathcal{C}}(-, -) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \longrightarrow \mathbf{An}.$$

Moreover, this functor preserves colimits separately in each variable.

Axiom The inclusion $\mathbf{An} \hookrightarrow \mathbf{Cat}_{\infty}$ admits both a left and a right adjoint (hence $\mathbf{An}$ has all limits and colimits). The left adjoint sends an ∞ -category $\mathcal{C}$ to its classifying anima $B\mathcal{C}$.

The right adjoint sends $\mathcal{C}$ to its maximal sub-anima $\mathcal{C}^{\simeq}$.

Def. Let $\mathcal{C}$ be an ∞ -category and $W \subset \mathcal{C}$ a subcategory (we think of W as specifying a collection of maps stable under composition). The localization $\mathcal{C}[W^{-1}]$ is the pushout of ∞ -categories

$$\begin{array}{ccc} W & \longrightarrow & \mathcal{C} \\ \downarrow & \ulcorner & \downarrow L \\ BW & \longrightarrow & \mathcal{C}[W^{-1}] \end{array}$$

By definition, $\mathcal{C}[W^{-1}]$ has the universal property that for any ∞ -category D , precomposition with L defines a fully faithful functor

$$-\circ L: \text{Fun}(\mathcal{C}[W^{-1}], D) \hookrightarrow \text{Fun}(\mathcal{C}, D)$$

with image those functors $\mathcal{C} \rightarrow D$ that send maps in W to equivalences. Axiom. There are functors $\mathbf{sSet} \rightarrow \mathbf{An}$ and $\mathbf{Top} \rightarrow \mathbf{An}$ that exhibit $\mathbf{An}$ as the localization (in the sense of ∞ -categories)

$$\mathbf{An} \simeq \mathbf{sSet}[W_{\text{he}}^{-1}] \simeq \mathbf{Top}[W_{\text{he}}^{-1}]$$

Notation For an ∞ -category $\mathcal{C}$, write $\mathcal{P}\mathcal{S}\mathcal{h}(\mathcal{C}) := \text{Fun}(\mathcal{C}^{\text{op}}, \mathcal{A}\mathcal{n})$.
and write $\mathcal{Y} : \mathcal{C} \rightarrow \mathcal{P}\mathcal{S}\mathcal{h}(\mathcal{C})$ for the functor $c \mapsto \text{Map}_{\mathcal{C}}(-, c)$.

Yoneda Lemma Let $\mathcal{C}$ be a small ∞ -category. For any $c \in \mathcal{C}$ and $F \in \mathcal{P}\mathcal{S}\mathcal{h}(\mathcal{C})$, there is a natural equivalence

$$\text{Map}_{\mathcal{P}\mathcal{S}\mathcal{h}(\mathcal{C})}(\mathcal{Y}(c), F) \xrightarrow{\sim} F(c)$$

Moreover, the functor $\mathcal{Y} : \mathcal{C} \rightarrow \mathcal{P}\mathcal{S}\mathcal{h}(\mathcal{C})$ is fully faithful.

Universal Property of presheaves. Let $\mathcal{C}$ be a small ∞ -category and let D be an ∞ -category with colimits. Then restriction along the Yoneda embedding defines an equivalence of ∞ -categories

$$\text{Fun}^{\text{colim}}(\text{Psh}(\mathcal{C}), D) \xrightarrow{\sim} \text{Fun}(\mathcal{C}, D)$$

colimit-preserving functors

$$[\bar{F}: \text{Psh}(\mathcal{C}) \rightarrow D] \longleftrightarrow [\bar{F} \circ \text{Yoneda}: \mathcal{C} \rightarrow D]$$

The inverse is given by sending a functor $F: \mathcal{C} \rightarrow D$ to its left Kan extension along the Yoneda embedding

$$\bar{F}(X) := \text{Colim}_{f(c) \rightarrow X} F(c)$$

Moreover, every colimit-preserving functor $\bar{F}: \text{Psh}(\mathcal{C}) \rightarrow D$ is a left adjoint with right adjoint $G: D \rightarrow \text{Psh}(\mathcal{C})$ given by

$$G(d)(c) = \text{Map}_D(F(c), d).$$

Monoids & Commutative monoids

> We can take Grothendieck's description of monoids as simplicial objects to define monoids in any ∞ -category.

Def. Let $\mathcal{C}$ be an ∞ -category with finite products. A monoid object of $\mathcal{C}$ is a functor $X: \Delta^{\text{op}} \rightarrow \mathcal{C}$ satisfying:

(1) $X_0 \xrightarrow{\sim} 1_{\mathcal{C}}$

(2) For each $n \geq 2$ and $i \in [n]$, the square

$$\begin{array}{ccc} X([n]) & \xrightarrow{\quad} & X(\{i < \dots < n\}) \\ \downarrow & & \downarrow \\ X(\{0 < \dots < i\}) & \xrightarrow{\quad} & X(\{i\}) \simeq 1_{\mathcal{C}} \end{array}$$

is a pullback.

We say that X is a group object if, additionally, the following stronger version of (2) is satisfied:

(3) For each $I \in \Delta$ and partition $I = S \cup T$ such that $S \cap T = \{i\}$ is a single element, the square

$$\begin{array}{ccc} X(I) & \longrightarrow & X(S) \\ \downarrow & & \downarrow \\ X(T) & \longrightarrow & X(\{i\}) \end{array}$$

is a pullback

The object X_1 is called the underlying object of X . We write $\text{Mon}(\mathcal{C}) \subset \text{Fun}(\Delta^{\text{op}}, \mathcal{C})$ for the full subcategory spanned by the monoids, and $\text{Grp}(\mathcal{C}) \subset \text{Mon}(\mathcal{C})$ for the full subcategory spanned by the groups.

Ex. Let $\mathcal{C}$ be an ∞ -category with finite limits and $x: 1_{\mathcal{C}} \rightarrow X$ a pointed object. The based loop object $\Omega_x X$ is the pullback

$$\begin{array}{ccc} \Omega_x X & \longrightarrow & 1_{\mathcal{C}} \\ \downarrow & \lrcorner & \downarrow x \\ 1_{\mathcal{C}} & \xrightarrow{x} & X \end{array}$$

To see why this definition is reasonable, consider $\mathcal{C} = \mathbf{A}^n$. Then elements of $\Omega_x X$ are supposed to be triples

$$(a \in *, b \in *, \text{an equivalence } x(a) \simeq x(b))$$

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$$(\text{an equivalence } x \simeq x)$$

That is, elements of $\Omega_x X$ are given by loops at x .

The object $\Omega_x X$ can be refined to a group object by taking the Čech nerve of the map $x \cdot 1_{\mathcal{C}} \rightarrow X$.

Given a map $f: Y \rightarrow X$, its Čech nerve is the simplicial object

$$\begin{array}{c} \cdots \quad \begin{array}{c} \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \end{array} \quad \begin{array}{c} Y \times Y \times Y \\ \times \quad \times \quad \times \end{array} \quad \begin{array}{c} \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \\ \rightrightarrows \end{array} \quad \begin{array}{c} Y \times Y \\ \times \end{array} \quad \begin{array}{c} \xrightarrow{\text{pr}_1} \\ \xleftarrow{\Delta} \\ \xrightarrow{\text{pr}_2} \end{array} \quad Y \quad \xrightarrow{f} \quad X \end{array}$$

has a natural augmentation

The Čech nerve always satisfies (2) above.

Def. A monoidal ∞ -category is a monoid in $\mathbf{Cat}_{\infty}$.

Remark. In order to formulate our more general version of toic geometry, we want to study the analogue of abelian groups in anima.

- > Recall the fundamental principle that we should not just remember that two entities are identified, but how they are identified.
- > So commutativity is not a condition, but an extra structure.
 - We're used to thinking of it as a condition because in 1-category theory the structure is unique if it exists.
 - This phenomenon already appears in the theory of symmetric monoidal 1-categories. The symmetry is not a property on a monoidal 1-category, but rather an additional structure one needs to put on a monoidal 1-category.

Def. Let $\mathcal{C}$ be an ∞ -category with finite products. A commutative monoid or $\mathbb{E}_\infty$ -monoid in $\mathcal{C}$ is a functor

$$X: \text{Set}_*^{\text{fin}} \longrightarrow \mathcal{C}$$

such that:

$$(1) X(*) \xrightarrow{\sim} 1_{\mathcal{C}}$$

(2) For each $n \geq 1$, the collapse maps

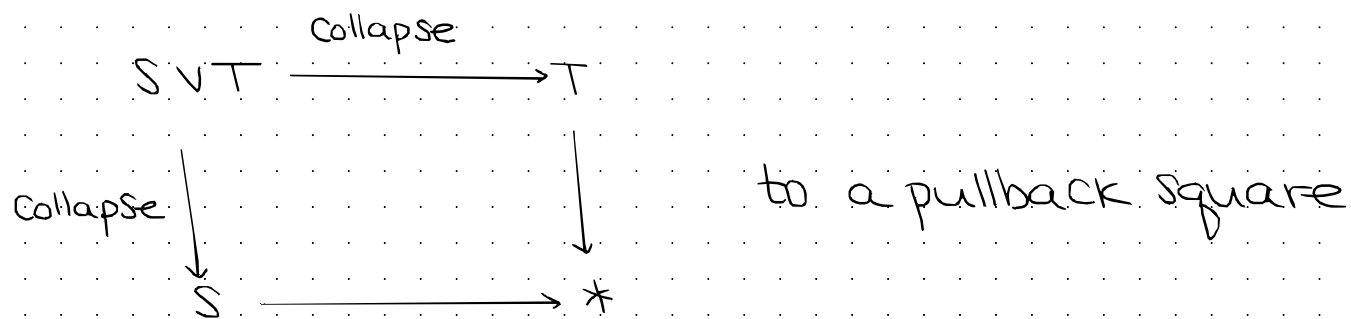
$$\begin{array}{ccc} \{1, \dots, n\}_+ & \longrightarrow & \{i\}_+ \\ j & \longmapsto & \begin{cases} i, & j=i \\ *, & j \neq i \end{cases} \end{array}$$

add disjoint basepoint $*$

induce an equivalence

$$X(\{1, \dots, n\}_+) \xrightarrow{\sim} \prod_{i=1}^n X(\{i\}_+)$$

Equivalently, for all finite pointed sets S and T , X sends the square



We write $\mathbf{CMon}(\mathcal{C}) \subset \mathbf{Fun}(\mathbf{Set}_*^{\text{fin}}, \mathcal{C})$ for the full subcategory spanned by the commutative monoids. We call $X(\{1\}_+)$ the underlying object of X .

Remark A monoid is often called an $\mathbb{E}_1$ -monoid. There is a whole hierarchy of $\mathbb{E}_n$ -monoids for $n \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$. The key example is that the n -fold loop object

$$\Omega_x^n X = \underbrace{\Omega(\cdots \Omega_x X)}_{n\text{-times}}$$

is an $\mathbb{E}_n$ -monoid (in fact, group). In ordinary category theory

- (1) A monoidal category is an $\mathbb{E}_1$ -monoid in the ∞ -category of 1-categories
- (2) A braided monoidal category is an $\mathbb{E}_2$ -monoid.
- (3) For $n \geq 3$ (including ∞) all the notions of an $\mathbb{E}_n$ -monoid agree, and an $\mathbb{E}_\infty$ -monoid in 1-categories is a symmetric monoidal category

We'll only need to work with monoids and commutative monoids (i.e., $n=1$ and $n=\infty$)

Exercise Consider the functor

$$\mathbf{CMon}(\mathbf{Set}) \longrightarrow \left\{ \begin{array}{l} \text{commutative monoids} \\ \text{in Set in the typical sense} \end{array} \right\}$$

given by $X \mapsto M = X(\{1\}_+)$ with unit

$$* = X(*) \longrightarrow X(\{1\}_+) = M$$

induced by the unique map $* \rightarrow \{1\}_+$ in $\mathbf{Set}_*^{\text{fin}}$ and multiplication.

$$M \times M \cong X(\{1, 2\}_+) \longrightarrow X(\{1\}_+) \cong M$$

induced by $1, 2 \mapsto 1$ and $*$ $\mapsto *$. Show that this functor is an equivalence of categories. The inverse sends $(M, m: M \times M \rightarrow M, 1 \in M)$ to

$$\{1, \dots, n\}_+ \mapsto M^{\{1, \dots, n\}}$$

with functoriality given by sending $\phi: \{1, \dots, m\}_+ \rightarrow \{1, \dots, n\}_+$ to the map

$$M^{\{1, \dots, m\}} \longrightarrow M^{\{1, \dots, n\}}$$

$$(x_i)_{i \in \{1, \dots, m\}} \mapsto \left(\prod_{i \in \phi^{-1}(j)} x_i \right)_{j \in \{1, \dots, n\}}$$

Here, the empty product is interpreted as the unit $1 \in M$.

> Let's explain how to compare monoids and commutative monoids

Definition Let $I \in \Delta$. The set of cuts of I is the quotient of the set of disjoint decompositions $I = S \sqcup T$ such that for all $s \in S$ and $t \in T$, we have $s < t$ where we identify $I \sqcup \emptyset$ and $\emptyset \sqcup I$.

The assignment $I \mapsto \text{Cut}(I)$ defines a functor

$$\text{Cut} : \Delta^{\text{op}} \longrightarrow \text{Set}_*^{\text{fin}}$$

with functionality given by sending $\phi : I \longrightarrow J$ to the map

$$\begin{aligned} \phi^* : \text{Cut}(J) &\longrightarrow \text{Cut}(I) \\ J = S \sqcup T &\longmapsto I = \phi^{-1}(S) \sqcup \phi^{-1}(T). \end{aligned}$$

Exercise. The functor Cut is isomorphic to the pointed simplicial set $\Delta^1 / \partial \Delta^1$. (This is the 'simplicial circle'.)

Lemma. Let $\mathcal{C}$ be an ∞ -category with finite products and $X \in \mathbf{CMon}(\mathcal{C})$. Then the functor

$$\Delta^{\text{op}} \xrightarrow{\text{cut}} \text{Set}_*^{\text{fin}} \longrightarrow \mathcal{C}$$

is a monoid object. Hence precomposition with cut defines a forgetful functor

$$-\circ \text{cut} : \mathbf{CMon}(\mathcal{C}) \longrightarrow \mathbf{Mon}(\mathcal{C}).$$

Def. A Symmetric monoidal ∞ -category is a commutative monoid in $\mathbf{Cat}_{\infty}$.

Observe. If $\mathcal{C}$ and $\mathcal{D}$ are ∞ -categories with finite products and $F : \mathcal{C} \rightarrow \mathcal{D}$ preserves finite products, then postcomposition with F defines functors

$$\mathbf{Mon}(\mathcal{C}) \longrightarrow \mathbf{Mon}(\mathcal{D}) \quad \text{and} \quad \mathbf{CMon}(\mathcal{C}) \longrightarrow \mathbf{CMon}(\mathcal{D})$$

compatible with the forgetful functors.

Example. Recall that the inclusion $\mathbf{An} \hookrightarrow \mathbf{Cat}_\infty$ preserves limits and admits a right adjoint $\mathcal{C} \mapsto \mathcal{C}^\sim$. Hence we obtain an adjunction

$$\mathbf{CMon}(\mathbf{An}) \rightleftarrows \mathbf{CMon}(\mathbf{Cat}_\infty)$$

That is, every symmetric monoidal ∞ -category $\mathcal{C}$ gives rise to a commutative monoid in $\mathbf{anima}$.

Ex. Consider the category $\mathbf{Set}^{\text{fin}}$ with symmetric monoidal structure given by disjoint union. Then

$$\mathbf{Set}^{\text{fin}}, \simeq \simeq \bigsqcup_{n \geq 0} B\Sigma_n$$

is a commutative monoid in $\mathbf{anima}$. One can show this is the free commutative monoid on a single generator.